Egyptian Informatics Journal 34 (2026) 100978

Contents lists available at ScienceDirect

Egyptian Informatics Journal

journal homepage: www.sciencedirect.com

Full Length Article

Alzheimer's detection using audio through deep learning techniques

Karthika Kuppusamy *, Anjana Rajesh

Department of Mathematics, Amrita School of Physical Sciences Coimbatore, Amrita Vishwa Vidyapeetham, India

ARTICLE INFO

Keywords:
Alzheimer'sdisorder
Speechanalysis
Temporal dynamics
BiLSTM
XGBoost

ABSTRACT

Alzheimer's Disease (AD) is a long-lasting neurodegenerative disorder that progressively weakens memory, communication, and cognitive abilities. Early detection and diagnosis are critical for timely intervention. Recently, Artificial Intelligence (AI) models with speech analysis have emerged in detecting AD using acoustic and prosodic characteristics of individuals. However, they still achieve lower accuracy due to a lack of temporal dynamics and complex nonlinear correlation features among the speech signals. Hence, a new robust, noninvasive, and cost-effective model is proposed in this article for early detection of AD from speech signals. A new hybrid model called TempoBoostNet is developed to simultaneously capture both temporal dynamics and complex nonlinear relationships among speech signals. First, a dataset of 547 speech recordings (247 CE and 300 Healthy Controls (HC)) is gathered from publicly available Kaggle and GitHub sources. The raw audio files are preprocessed using the Wiener filter and Mahalanobis distance to eliminate noise and silence, respectively. Then, various features that reflect both low-level acoustic characteristics and higher-level speech dynamics are extracted. Also, contextual embeddings from Wav2Vec2.0 and Perceptual Linear Prediction (PLP) coefficients are extracted. All extracted features are concatenated to form a unified multi-level feature set for better feature representation. Moreover, this feature set is utilized to train the TempoBoostNet, which is built by hybridizing the Bidirectional Long Short-Term Memory and Extreme Gradient Boosting (BiLSTM-XGBoost) classifier for AD detection. Finally, experimental results show that this TempoBoostNet achieves 97.4% accuracy, 97.4% precision, 97.3% recall, and 97.3% F1-score, outperforming traditional models.

1. Introduction

AD is one of the most predominant neurodegenerative disorders among the aged. Statistics from the World Health Organization (WHO) indicate that the prevalence of AD has exhibited a notable increase recently, with projections estimating that the patient counts can reach 152 million by 2050, thereby posing a serious threat to public health and imposing a substantial economic burden on home care and social security systems. The advancement of AD begins with imperceptible cerebral alterations in the person and culminates in memory impairment and subsequent physical incapacity [1]. Medical therapies for AD are currently restricted to alleviating clinical manifestations, without delivering therapeutic or defensive influences on the primary procedures [2]. While memory loss and cognitive decline are important markers of AD, language impairment is also a significant symptom that aids in identifying the illness and evaluating its severity. Speech may elucidate an individual's cognitive condition and disclose more dimensions of cerebral impairment [3].

Previous studies have emphasized the benefits of utilizing speech as a biomarker for AD detection. They indicate that language impairments often manifest early in the progression of AD, and it is feasible to identify by analyzing the acoustic and linguistic characteristics of individuals [4]. Certain research has presented findings on identifying features between the AD group and the HC group. Recent developments in AI, predominantly in the fields of voice classification [5] and Natural Language Processing (NLP) [6], offer hopeful opportunities for AD diagnosis [7–12]. To detect AD based on the acoustic data, Tripathi and Kumar [13] explored a variety of Machine Learning (ML) methods, including K-Nearest Neighbor (KNN), Decision Tree (DT), Support Vector Machine (SVM), XGBoost, and Random Forest (RF). However, they obtained an overall accuracy of 75.59% using the XGBoost classifier. To detect AD, an ensemble ML method was developed in [14] by combining KNN, RF, DT, SVM, and Multi-Layer Perceptron (MLP). This approach focused on identifying anomalies and determining whether a patient has AD by utilizing attributes extracted from speech signals. Despite achieving 97% accuracy, this approach is unable to capture temporal and complex

* Corresponding author.
E-mail addresses: k_karthika@cb.amrita.edu (K. Kuppusamy), cb.ps.p2asd23030@cb.students.amrita.edu (A. Rajesh).

https://doi.org/10.1016/j.eij.2026.100978
Received 19 January 2026; Received in revised form 15 March 2026; Accepted 21 April 2026
Available online 6 May 2026
1110-8665/© 2026 THE AUTHORS. Published by Elsevier BV on behalf of Faculty of Computers and Artificial Intelligence, Cairo University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

properties. In [15], a Deep Learning (DL) technique, such as ResNet50, is employed to identify AD by considering various speech features, including raw phase, modulated group delay, instantaneous frequency phase, and magnitude spectrograms. However, due to its inability to capture temporal fluctuations in voice signals, it still achieved an accuracy of 90%. Thus, although several ML and DL models have evolved for AD diagnosis, they still have poorer accuracy due to a lack of capturing complex temporal dynamics and minor prosodic changes in speech data.

This demands a robust, low-cost, speech-driven model that makes use of both sequential dependencies and non-linear feature interactions to detect AD early and reliably. Therefore, this article introduces a new model by hybridizing both ML and DL models for accurate, non-invasive, and cost-effective early detection of AD from speech signals. This study aims to increase the accuracy of AD detection by capturing both temporal dynamics and complex non-linear patterns characteristic of speech signals. The primary contributions of this work are.

1. Applying a new audio preprocessing approach by combining Wiener filtering and Mahalanobis distance for eliminating noise and silence, respectively, from the speech signals of AD and healthy individuals.
2. Extracting and concatenating various acoustic and prosodic features with deep contextual embeddings from pre-trained Wav2Vec 2.0 to create a multi-level feature set for AD detection.
3. Developing the TempoBoostNet model by integrating BiLSTM and XGBoost classifier to capture long-range temporal dependencies and non-linear relationships from speech signals, respectively, to classify AD and HC individuals precisely.
4. Evaluating the effectiveness of this TempoBoostNet with conventional ML and DL models like SVM, RF, DT, etc., to comprehend its success in AD detection.

The remaining sections are prepared as follows: Section 2 debates earlier work. Section 3 explains the TempoBoostNet model for AD detection. Section 4 reveals test outcomes. Section 5 summarizes this work.

2. Literature review

Recently, there has been an increased interest in speech-based biomarkers in AD due to the non-invasive and inexpensive characteristics of voice analysis. Several studies have already investigated a wide variety of acoustic, prosodic, and linguistic characteristics in combination with ML and DL approaches. This section provides a review of speech-based detection of AD and related applications to other health-related conditions.

2.1. Review on Speech-Based detection for Alzheimer's disease

Mahajan and Baths [16] developed the Convolutional Neural Network (CNN)-LSTM model to identify AD from spontaneous speech. They explored the DL model by integrating acoustic features into a common vector utilizing recurrent units. This model achieved an accuracy of 72.92%. Language-agnostic voice characteristics like word-level length, pause rate, and speech intelligibility were examined by Shah et al. [17]. They achieved 85% accuracy by using Logistic Regression (LR) for classification and Principal Component Analysis (PCA) for dimensionality reduction. Chen et al. [18] described a cross-lingual AD detection scheme using speech characteristics. They achieved 69.6% accuracy using SVM with a linked layer. Shah et al. [19] subjected the main components obtained from *meta*-traits, word-level period, pause rate, and voice intelligibility to LR with L2 regularization, resulting in an accuracy of 69.57%. Burke et al. [20] achieved a classification accuracy of 78.4% by using LR on both newly generated lexical-semantic features and Ostrand and Gunstad's lexical-semantic features. In [21], the authors utilized Vision Transformer (ViT) for audio spectrogram images and BERT for transcripts to identify AD with 91.25% accuracy.

Ahn et al. [22] explored different DL models, such as Inceptionv3, VGG19, Xception, ResNet50, and DenseNet121, for detecting AD from Mel-Frequency Cepstral Coefficient (MFCC) features of speech signals. Among these models, DenseNet121 achieved an accuracy of 90%. Park et al. [23] utilized the Deep Neural Network (DNN) to predict AD dementia from voice features such as MFCCs and chroma. This model achieved an accuracy of approximately 82%. Xu et al. [24] developed a Pre-trained Ensemble-based AD Detection (PEADD) using ResNet, VGG, and EfficientNet as base learners for voice-based AD detection, with an accuracy of 92%. It limited generalizability because the considered dataset may not fully reflect the characteristics of AD. Priyanka and Amshakala [25] predicted dementia via audio using ensemble and DL models from acoustic features. They extracted voice features and fed them into the ensemble model, such as RF, AdaBoost, XGBoost, and gradient boost. They also used DL models such as BiLSTM, LSTM, and CNN-LSTM for AD detection. They observed that gradient boost attained an accuracy of 90.5%.

2.2. Review on Speech-Based detection for other health conditions

Kadhim et al. [26] presented a new voice categorization method based on Gower distance for detecting Parkinson's disease. Additionally, they utilized the cuckoo search scheme for feature dimensionality reduction, achieving an accuracy of 98.3%. Yuan et al. [27] used language signal features to train an ensemble ML classifier, including LR, RF, KNN, and DNN as base learners for Parkinson's disease detection, achieving an accuracy of 95%.

Alghamdi et al. [28] explored MFCC traits and trained a deep CNN to accurately detect respiratory disorders with an accuracy of 95.1%. In [29], a new triplet Time-Frequency Representation (TFR)-driven Multi-Head Self-organized Operational Neural Network (MHSONN) was developed for chronic obstructive pulmonary disease detection using lung sounds, achieving an accuracy of 99.81%. Pradhan et al. [30] created a cascaded Perceptual Functional Link Artificial Neural Network (PELANN) model for respiratory disease detection with an accuracy range between 92–95% for different datasets. In [31], a bilingual dual-head deep neural structure was presented for detecting Parkinson's disease from speech signals with an accuracy of 90.83%.

2.3. Research Gap

Despite the exploration of various ML and DL models for detecting Alzheimer's speech, many current approaches exhibit significant limitations. Research often relies on small feature sets that cannot capture temporal dynamics and complex interactions in speech. While models such as CNNs, LSTMs, and ResNet can process sequential data, they often struggle with issues like overfitting and inadequate representation of acoustic features. Ensemble ML methods, although moderately accurate, fail to address temporal nuances and contextual elements in voice data. As a result, a robust TempoBoostNet model is developed in this study to enhance the accuracy of detecting AD by modeling temporal and non-linear relationships between speech signals.

Table 1 summarizes the models studied above in terms of their advantages, disadvantages, datasets used, and the values of adopted metrics.

3. Proposed Methodology

This section briefly describes the TempoBoostNet model for AD detection. Fig. 1 illustrates the schematic pipeline of this study.

3.1. Dataset

The speech corpus used in this work was created by collecting audio files from two publicly available repositories: the Federated Alzheimer

Table 1
Summary of Literature Review on Speech-Based AD Detection and Related Health Conditions.

Ref. No.	Models	Dataset Used	Accuracy	Merits	Demerits
[16]	CNN-LSTM	DementiaBank	72.92%	It combines acoustic features with recurrent units for sequential modeling, achieving integrated speech representation.	Struggles to capture complex temporal and prosodic features effectively.
[17]	LR + PCA	DementiaBank	85.00%	It uses language-agnostic features like pause rate and speech intelligibility, making it computationally lightweight.	Limited feature set fails to capture nonlinear or deep acoustic patterns.
[18]	SVM with linked layer	ADReSS-M challenge dataset	69.57%	Cross-lingual approach broadens applicability across languages and utilizes paralinguistic features.	Cannot model complex temporal speech dynamics effectively.
[19]	LR with L2 regularization	DementiaBank	69.57%	Regularization reduces overfitting and *meta*-trait and pause features provide linguistically meaningful signals.	Limited to surface-level linguistic features without deep temporal analysis.
[20]	LR	DementiaBank Pitt Corpus	78.40%	It combines novel lexical-semantic features with existing ones for a richer speech representation.	Relies solely on lexical features without integrating acoustic or prosodic cues.
[21]	BERT + ViT + Crossmodal Attention	ADReSS-M challenge dataset	91.25%	Multimodal fusion of audio spectrograms and text transcripts captures complementary speech information.	Requires transcripts alongside audio, increasing complexity and limiting purely speech-based deployment.
[22]	DenseNet121	MFCC-based speech data	90.00%	It learns hierarchical representations.	Feature input limited to MFCCs, lacking temporal modeling for capturing speech sequence dynamics.
[23]	DNN	DementiaBank Pitt Corpus	82.00%	It captures spectral features relevant for AD-related vocal changes.	Lacks sequential modeling capability for temporal speech dependencies.
[24]	PEADD	Audio-based AD dataset	92.00%	Ensemble of strong pretrained base learners improves generalization.	Limited generalizability as the dataset may not represent the full diversity of the AD population.
[25]	Gradient boost	Audio-based dementia dataset	90.50%	It has high robustness and flexibility.	The accuracy is still low.
[26]	Gower Distance + Cuckoo Search	Parkinson's disease voice dataset	98.30%	Novel Gower distance metric for voice classification; cuckoo search reduces dimensionality effectively.	Tailored for Parkinson's disease; may not generalize to AD or other neurodegenerative speech disorders.
[27]	Ensemble	Parkinson's speech signal dataset	95.00%	Ensemble of diverse classifiers improves robustness.	Limited applicability to cognitive disorders like AD.
[28]	Deep CNN with MFCC	Respiratory sound dataset	95.10%	MFCC features combined with deep CNN; effective for audio-based disease detection.	Specific to respiratory disorders; MFCC-only input may limit the detection of subtle speech pattern changes.
[29]	Triplet TFR + MHSONN	Lung sound dataset	99.81%	Multi-head self-organized neural network achieves near-perfect accuracy for COPD detection.	Highly specialized for COPD; complex architecture may be difficult to adapt for AD speech analysis.
[30]	Cascaded PELANN	Respiratory disease dataset	92–95%	Cascaded architecture with bio-inspired computation yields strong accuracy across multiple datasets.	Focused on respiratory diseases; performance may vary significantly across diverse speech datasets for AD.
[31]	Bilingual Dual-Head DNN	Parkinson's disease speech dataset	90.83%	Bilingual approach extends applicability; dual-head architecture separates language-specific speech patterns.	Limited to Parkinson's disease; dual-head design adds training complexity.

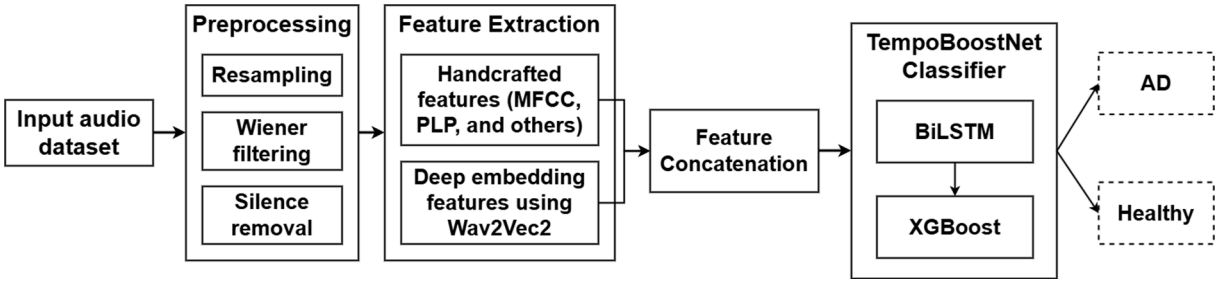


Fig. 1. Schematic Pipeline of this Proposed Study.

Speech Dataset on Kaggle [32] and the Alzheimer Speech Corpus on GitHub [33]. This dataset comprises a total of 547 audio recordings (247 audio samples of patients with AD and 300 audio samples of HC). Each recording corresponds to an independent speech sample from different speakers, ensuring speaker independence in the dataset. Of these, 80% are used for training and 20% are used for testing. The majority of the recordings consist of spontaneous speech, picture-description activities, and brief narrative replies, with a smaller portion consisting of sustained vowel sounds and prompted sentences, depending on the specifications of the sources. Although demographic metadata (e.g., gender, age distribution, educational attainment) is not equally available in both datasets, most of the recordings are from older adults who speak English, with the majority being 60 years old and above, fitting the target demographic for AD screening.

The recordings were made in various, mostly low-noise conditions using standard microphones from personal devices, clinical equipment, or telephonic communication systems. The sampling rates are 16 kHz and 44.1 kHz. Due to the real-world settings in which the datasets were recorded, there is natural heterogeneity in speech quality, background noise, and microphone behavior, allowing for the creation of models that can be well generalized to realistic diagnostic settings. There is no additional metadata available in the public repositories (e.g., medication status, disease severity indices, formal cognitive assessment). After collecting the dataset, the audio files were standardized through

resampling, noise removal, and silence removal, followed by feature extraction to ensure consistency and comparability across the sample set.

Despite the fact that the dataset utilized in the current study includes 547 speech recordings, several steps have been taken to minimize overfitting and enhance model generalization. Regularization techniques, such as dropout, L2 kernel regularization, and XGBoost classifier with shrinkage and tree pruning, have been employed to control the complexity of the model and improve generalization results. Additionally, the dataset was obtained from real-life sources, which had natural fluctuations in recording conditions, further contributing to the strength of the proposed model.

3.2. Data preprocessing

Enhancement in audio quality is necessary for applications such as speech processing, medical diagnosis, and ML. This phase of the work employs techniques for noise and silence removal as a preprocessing stage in the classification of AD. Using the Mahalanobis distance helps eliminate silence, and Wiener filtering is used to reduce noise in the audio. The processed audio is then subjected to feature extraction and fed as input to a model that evaluates the person for AD.

3.2.1. Noise removal using Wiener filtering

The Wiener filter is an optimal technique for signal processing to reduce the error between the estimated and original signals. It is widely used for noise suppression. The equation for the frequency response of a Wiener filter is given as:

$$H(f) = \frac{s_{xx}(f)}{s_{xx}(f) + s_{nn}(f)} \tag{1}$$

In Eq. (1), $s_{xx}(f)$ and $s_{nn}(f)$ are the power spectral density of the actual signal and noise, respectively. In this study, Wiener filtering is applied to smooth the signal while preserving speech characteristics.

3.2.2. Silence removal using Mahalanobis distance

Silence removal is a crucial step to improve speech clarity and efficiency in speech analysis tasks. The Mahalanobis distance method is used to classify frames as either voiced or silent based on statistical properties. The equation for Mahalanobis distance is given as:

$$D_M(x) = \sqrt{(x-\mu)^T S^{-1}(x-\mu)} \tag{2}$$

In Eq. (2), x is the feature vector, μ defines the mean, and S indicates the covariance matrix of the feature distribution. The steps for removing silence speech signals are as follows:

- Estimate background noise statistics for every 200 ms.
- Calculate the background noise for each frame.
- Categorize frames exceeding a predefined threshold as voiced.
- Categorize all frames not exceeding the threshold as unvoiced.
- Consider frames with less sound activity (e.g. silence or low energy) as silenced.

3.3. Feature extraction

This study employs a comprehensive audio feature extraction pipeline to extract a wide variety of acoustic, prosodic, and learned speech representations to detect patterns linked with AD.

3.3.1. MFCC

The MFCCs are the key spectral features in speech processing that imitate human hearing, wherein perceptually relevant frequency bands are enhanced to capture fine articulatory gestures with high fidelity. Fig. 2 shows the MFCC extraction steps.

First, the speech signal $s(t)$ is split into short overlapping frames with a usual 50% overlap, with each frame $sm(t)$ defined for $t \in [mT, (m+1)T]$. The Hamming window $w(t)$ reduces the edge discontinuities, and the equation for it is given by

$$w(t) = 0.54 - 0.46\cos\left(\frac{2\pi t}{N-1}\right) \tag{3}$$

Second, for all frames, the Fast Fourier Transform (FFT) is computed to yield the frequency spectrum as $S_m(f) = F\{S_m(f)\}$. The output spectrum goes through Mel scale filter banks to achieve a simulation of the human ear's non-linear frequency perception, where the Mel scale is defined as:

$$Mel(f) = 1127\log\left(1 + \frac{f}{700}\right) \tag{4}$$

Then, the energy for each Mel band is calculated as:

$$M_m(k) = \sum_f |S_m(f)|^2 . H_k(f) \tag{5}$$

Followed by the logarithmic transformation of Mel-filtered energies,

$$L_m(k) = \log(M_m(k)) \tag{6}$$

To simulate loudness, the last step involves applying the Discrete Cosine Transform (DCT) to the log Mel spectrum, resulting in the MFCC coefficients.

$$C_m(n) = \sum_{k=0}^{K-1} L_m(k) \times \cos\left(\frac{\pi}{K}\left(n + \frac{1}{2}\right)k\right) \tag{7}$$

In Eq. (7), K represents the number of Mel bands, typically set to 12 or 13.

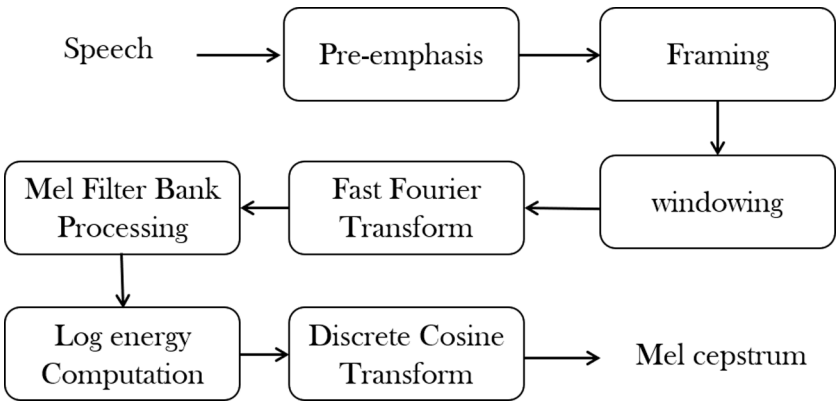


Fig. 2. Extraction of MFCC Features.

3.3.2. PLP features

The PLP coefficients, scientifically established through psychoacoustic principles applied to traditional LPC, bring validity to human auditory perception by transforming the speech signal $x(t)$ to its frequency domain using the FFT:

$$X(f) = \mathscr{F}\{x(t)\} \tag{8}$$

Fig. 3 illustrates the processes involved in extracting PLP features. A frequency spectrum is filtered by an array of bandpass filters, $H_b(f)$, which mimic the critical bands of the human ear and are designed to approximate its important frequency bands, typically defined in relation to the Bark scale. The energy in each frequency band E_b is:

$$E_b = |X(f)|^2 . H_b(f) \tag{9}$$

Then, it is followed by applying equal loudness preemphasis, which, in essence, compensates for the different frequency sensitivity of the human ear. This means enhancing the mid frequencies while suppressing the extremes of this spectrum to simulate human hearing based on equal loudness contours. The intensity loudness power law mimics the logarithmic characteristic of human ear sensitivity to different sound intensities. Next, the energy from each frequency band undergoes a power transformation defined as follows:

$$L_b = \log(E_b^{\gamma}) \tag{10}$$

In Eq. (10), L_b indicates the energy in the b band log transformed and γ is often set in the range of 0.3—0.5 to represent loudness perception. These are then followed by psychoacoustic mapping conversions using Linear Predictive Coding (LPC). The goal of LPC is to minimize the prediction error $e(t)$ between the input speech signal $x(t)$ and the predicted signal $\hat{x}(t)$:

$$e(t) = x(t) - \hat{x}(t) = x(t) - \sum_{k=1}^{p} a_k x(t-k) \tag{11}$$

In Eq. (11), $\hat{x}(t)$ denotes the predicted speech sample, a_k contains the LPC coefficients, and p describes the order of the LPC analysis. The coefficients a_k are found determined by minimizing the error between the actual and predicted signal, representing the spectral envelope of the speech signal. Finally, the PLP coefficients are derived by inverting the LPC coefficients from the transformed signal.

3.3.3. Other features related to speech signal

In addition to MFCC and PLP, the following features are extracted for AD detection:

- Chroma features are computed by projecting the whole spectrum onto 12 pitch classes (one for each semitone of the musical octave) without any regard for the octave, to concentrate on harmonic and melodic properties. The chroma vectors are typically drawn using either a Short Time Fourier Transform (STFT) or Constant Q Transform (CQT) and help determine pitch stability and tonal control parameters that may be gradually negatively affected by CI.
- Spectral contrast is referred to as the variance in energy between the spectral peaks and the valleys in sub-bands of the frequency spectrum and is responsible for capturing the texture and timbre of speech. The spectral contrast is thus determined by partitioning the spectrum into different frequency bands and finding the contrast between the highest and lowest magnitudes on each band. Variations in spectral contrast invariably correlate with changes in voice quality and breathiness, phenomena typical in Alzheimer's speech.
- Jitter and shimmer are measured using Praat's implementation of autocorrelation and peak picking methods for measuring micro variations in F0 and amplitude, respectively. Jitter quantifies the pitch variation from cycle to cycle, whereas shimmer quantifies the amplitude variation from cycle to cycle. Algorithms predicting jitter commonly involve pitch period extraction and then computing the relative period differences; on the other hand, shimmer depends on the analysis of the intensity envelope. High jitter and shimmer displace the voice, which is reduced due to neuromuscular degeneration in AD.
- The Zero Crossing Rate (ZCR) is the rate at which the signal deviates through frames and is computed by counting the number of times the signal amplitude crosses zero in a particular frame. ZCR provides information about the noisiness of the signal; greater noisiness and irregularity in articulation can be associated with cognitive decline.
- Spectral bandwidth, calculated as the second central moment of the power spectrum, is a measure of the spread of frequencies about the spectral centroid, thus capturing characteristics of articulation and resonance.
- Pause length is measured through silence segmentation algorithms that detect silent regions by applying thresholds on short-term energy or spectral entropy. Longer and more frequent pauses, as quantified by the mean and distribution of silent intervals, indicate some word retrieval problems and cognitive load during spontaneous speech.
- Pitch statistics (F0) are accomplished with the use of either autocorrelation or cepstral methods for precise pitch tracking. Mean F0, F0 variance, and F0 range are given for considerations on prosodic flattening or emotional blunting: all affective states that characterize Alzheimer's patients, particularly in the early stage, who also demonstrate later functional impairment, exhibiting memory.
- Log Mel spectrograms are generated with a Mel-scaled filterbank applied to the power spectrum before logarithmic compression. This has the quality of perceiving the audio well and then ensuring the most critical frequencies are aligned with an important perceptual aspect in accordance with human auditory perception.

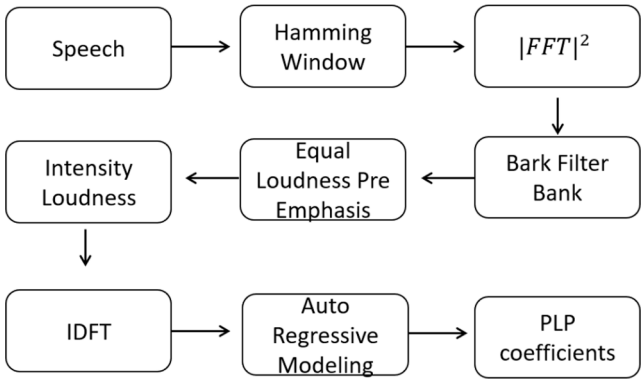


Fig. 3. Processes in PLP Feature Extraction.

3.3.4. Feature extraction using Wav2Vec2

This study utilized the Wav2Vec2.0 model, specifically the base model trained on LibriSpeech, to extract speech features. The embeddings were derived from the final transformer layer, representing each audio segment. The audio inputs were typically resampled to 16 kHz and scaled to values between 0 and 1 before feature extraction to meet the model's input requirements. The Wav2Vec 2.0 architecture generates embeddings of varying lengths, corresponding to temporal frames of the audio signal. Mean pooling was applied in the temporal dimension to create a single 768-dimensional vector per audio sample, converting them into a fixed-size representation for further processing.

Table 2 presents an evaluation between MFCC and PLP features on several classical ML algorithms (SVM, RF, LR, and DT). The Alzheimer's speech corpus described in Section 3.1 was used in the baseline experiments, which employed a uniform preprocessing pipeline including Wiener filtering, Mahalanobis-based silent removal, and uniform feature extraction. This shows the discriminative ability of these two popular

Table 2
Comparison of MFCC and PLP Features for Various Models.

Model	MFCC Accuracy	PLP Accuracy
SVM	70.3%	73.1%
RF	75.2%	74.4%
LR	69.5%	70.3%
DT	67.6%	67.2%

handcrafted feature sets individually, before their integration into the multi-level feature framework discussed in this paper. This comparison underscores the limitations of using either MFCC or PLP features alone, emphasizing the benefits of the more comprehensive hybrid feature-fusion implemented within the proposed TempoBoostNet framework.

3.3.5. *Feature Concatenation*

After extracting MFCC, PLP, chroma, spectral contrast, jitter, shimmer, pitch statistics, pause-related prosodic features, and log-Mel spectrograms, as well as high-level contextual embeddings based on Wav2Vec 2.0, all features are then fused in a single feature fusion pipeline and passed into the BiLSTM.

Initially, the groups of features are standardized through z-score normalization to ensure that features with a large difference in their numbers have an equal effect on the training process. Frame-level signals (i.e., MFCCs, PLPs, chroma, jitter, shimmer, and F0 statistics) are adapted in time by dividing each audio signal into fixed-size overlapping frames (a 25-ms window with a 10-ms hop). At each temporal frame, the respective feature vectors are pooled to produce a high-dimensional composite feature that captures spectral, prosodic, and articulatory features simultaneously. Since Wav2Vec 2.0 provides contextual embeddings at a different time resolution, these features are resampled by linear interpolation, either by up-sampling or down-sampling, to match their frame rate with that of the handcrafted features. Once all timesteps are aligned, the Wav2Vec 2.0 embedding of the time step is added to the handcrafted feature vector to obtain the final multi-level fused representation.

The final series of concatenated feature vectors is then input into the BiLSTM layer. This fusion strategy ensures that the BiLSTM receives a well-endowed representation that is temporally synchronized and capable of capturing long-range temporal dependencies as well as nonlinear speech variations caused by AD. The BiLSTM network is a stacked network that has 128 hidden units per directional stream in two layers. The fused feature sequence is the length of the sequence of temporal frames, which forms the input sequence. The BiLSTM produces a series of hidden states at each time step, which are then summed up across the time dimension to produce a fixed-size vector of dimensionality 256 (i.e., 128 units per direction). This vector is then used to identify AD by forwarding it to an XGBoost classifier.

Transitioning from a deep sequential model (BiLSTM) to a tree-based ensemble classifier (XGBoost) is chosen over an end-to-end DL with a softmax layer due to limitations in prior speech-based AD detection studies. BiLSTM excels at modeling temporal dependencies but can overfit on small datasets. XGBoost's regularization and effectiveness on structured inputs make it a better choice for AD detection with limited samples. Aggregating BiLSTM hidden states and feeding them into XGBoost allows for learning complex decision boundaries. This hybrid approach combines representation and discriminative learning without the instability of joint training.

3.4. *Proposed TempoBoostNet model for AD detection*

In this research work, a hybrid learning model called TempoBoostNet is introduced, which uses a combination of ML and DL approaches to classify AD based on features from speech (Fig. 4). The model is created by merging the BiLSTM network with an XGBoost classifier to exploit their advantages: temporal modeling and robust decision-making. Cognitive decline that occurs from a disease such as Alzheimer's changes how a person talks over time. In fact, changes may be identified as characteristic patterns in speech, such as pauses or difficulties producing sentences. Tracking changes in speech over time will provide a way to identify possible periods, or 'episodes,' of reduced cognitive output. Disfluencies, longer pauses, flattened prosody, and irregularities in the rhythm of speech are some of the characteristics of speech behavior typical of an Alzheimer's patient.

BiLSTM networks could have captured such patterns well due to their treatment of input in past and future sequences. Here, the BiLSTM is associated with various speech features such as MFCCs, chroma, pitch, jitter and shimmer, pause duration, PLP, and wav2vec embeddings, enabling it to learn long-range dependencies and fluctuations in the context of speech. For these reasons, harnessing strategies for regularization is quite critical in highly overparameterized deep networks. L_2 kernel regularization ($\lambda = 0.02$) serves mainly as a deterrent to overfitting. This is achieved by introducing a penalty into the loss function.

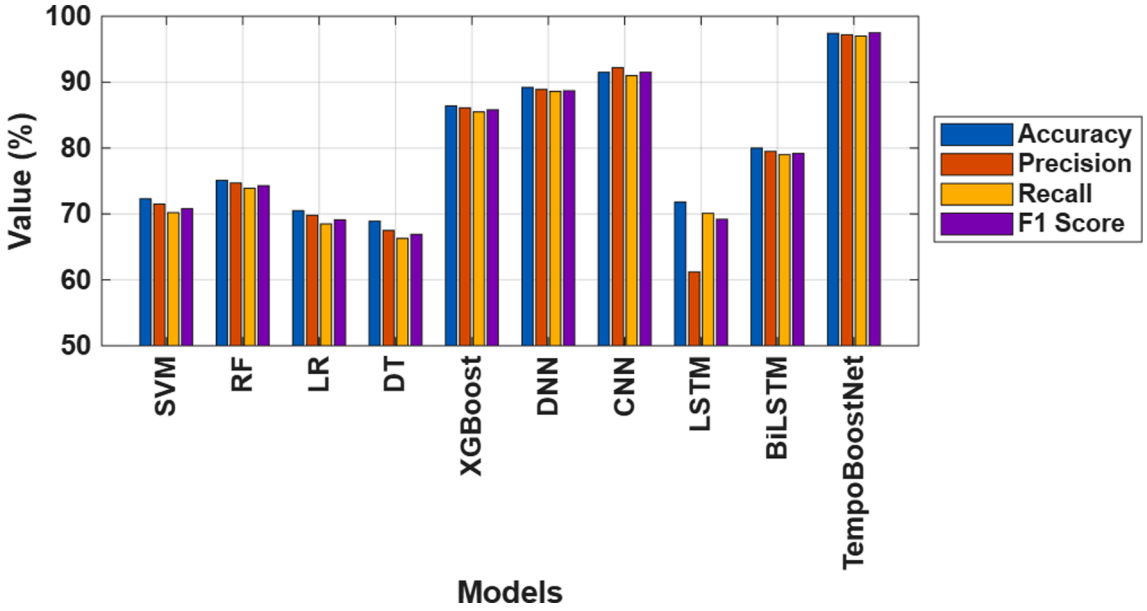


Fig. 4. Comparison of Proposed TempoBoostNet and Existing Models for AD Detection.

The modified loss function would take the following form:

$$L_{total} = L_{original} + \lambda \sum_i w_i^2 \quad (12)$$

In Eq. (12), w_i represents the model weights. This model incorporates a dropout probability of 0.6 to improve robustness by randomly deactivating sections of the network during training. A dense layer with 32 units and a 'swish' activation function refines the output of the BiLSTM. The Adam optimizer is utilized for training, with gradient clipping (clip norm of 1.0) to ensure stable updates and address the issue of exploding gradients. Instead of a softmax layer, the learned features from the BiLSTM are inputted into an XGBoost classifier for classification.

The XGBoost classifier is a robust gradient boosting tree ensemble method recognized for its effectiveness on tabular and structured data. It enhances classification by incorporating additional learning stages based on the rich sequential representations generated by the BiLSTM. To prevent overfitting, regularization techniques such as shrinkage and tree pruning are applied to enhance the model's generalization to unseen data.

Tree pruning eliminates unnecessary branches from decision trees, simplifying the model and reducing sensitivity to noise in the training data. Shrinkage, or learning rate, determines the impact of each tree on the model, with updates at each boosting step calculated as:

$$F_m(x) = F_{m-1}(x) + \eta \times h_m(x) \quad (13)$$

In Eq. (13), $F_m(x)$ represents the ensemble model after m iterations, $h_m(x)$ is the new weak learner, and η is the learning rate that controls the contribution of each tree. Therefore, learning should be gradual to reduce the model's vulnerability to fitting random fluctuations in the data.

4. Results and discussion

This section assesses the efficacy of TempoBoostNet against baseline and existing AD detection models using the dataset described in Section 3.1. The tests are carried out on a Windows 10 OS-powered system with an Intel Core i7, 16 GB of RAM, and an NVIDIA GTX-1060 GPU. The hyperparameters of the proposed TempoBoostNet model were selected through an experimental tuning process. Different values of parameters such as the number of BiLSTM layers, hidden units, dropout rate, learning rate, batch size, and XGBoost parameters were tested during the training phase. The parameter combination that provided stable training performance and improved classification results was selected. The final hyperparameter settings used in this study are presented in Table 3.

To prevent possible leakage of the data, the dataset was strictly split into independent training (80%) and testing (20%) sets before training the models. The entire process of preprocessing, feature extraction, and model training was done using the training data only, with the test set being used only for final evaluation. This segregation ensures that the model does not have access to test samples during the learning process.

To evaluate the proposed and existing AD detection models, the following metrics are considered:

- **Accuracy:** It represents the overall correctness of the model evaluated by comparing the correctly predicted observations to the total observations.

$$Accuracy = \frac{TruePositive(TP) + TrueNegative(TN)}{TP + TN + FalsePositive(FP) + FalseNegative(FN)} \quad (14)$$

- **Precision:** It is used to measure the ratio of positive identifications that were actually correct. It is defined by

$$Precision = \frac{TP}{TP + FP} \quad (15)$$

- **Recall:** It measures the number of actual positives correctly pinpointed by the model. It is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (16)$$

- **F1-Score:** It is determined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (17)$$

Table 4 and Fig. 4 present the performance evaluation of the proposed TempoBoostNet model compared to baseline models such as SVM, RF, LR, DT, XGBoost, DNN, CNN, LSTM, and BiLSTM. The results suggest that classical classifiers and DL techniques struggle to effectively handle the spectral and temporal complexities present in speech data. While neural models like CNN, LSTM, and BiLSTM consistently outperform SVM, RF, LR, and DT across all metrics, they do not perform exceptionally well. The high accuracy achieved by TempoBoostNet can be attributed to the combination of temporal feature learning through BiLSTM, strong nonlinear classification with XGBoost, and multi-level feature representation extracted from speech signals.

These results corroborate the assertion that the introduced time-dimensional models enhance the robustness of voice-based detection of AD. A hybrid model was used, which combined the robust decision-making of XGBoost with the temporal feature extraction capabilities of Bi-LSTM to further boost classification efficacy. The BiLSTM-XGBoost model performs noticeably better than every previous model. It routinely obtains good precision, recall, and F1-score values, and it

Table 3
Hyperparameter Settings.

Model	Hyperparameters	Values
BiLSTM	No. of layers	2
	No. of hidden units	128
	Dropout rate	0.6
	Recurrent dropout	0.3
	L_2 kernel regularization	0.02
	Optimizer	Adam
	Learning rate	0.001
	Epochs	100
	Batch dimension	32
XGBoost	No. of estimators	500
	Maximum depth	6
	Subsample	0.8
	Learning rate	0.05

Table 4
Results of Evaluation Metrics for Proposed and Existing Models for AD Detection.

Models	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
SVM	72.1	70.0	71.0	72.3
RF	75.0	74.8	74.9	75.1
LR	70.0	69.0	69.5	70.5
DT	68.0	67.5	67.8	68.9
XGBoost	86.0	86.5	86.3	86.5
DNN	89.1	89.0	89.1	89.2
CNN	91.7	91.2	91.4	91.5
LSTM	61.8	70.0	65.6	72.4
BiLSTM	79.9	79.8	79.8	80.0
Proposed TempoBoostNet	**97.4**	**97.3**	**97.4**	**97.4**

achieves an impressive accuracy of 97.4%. This significant gain demonstrates how combining gradient boosted DTs with sequential learning can enable the model to capture complicated non-linear relationships as well as temporal dependencies in the data. The outcomes demonstrate how well the suggested method works to identify speech patterns associated with AD reliably.

The Confidence Intervals (CIs) can be seen in Table 5 and clearly indicate that TempoBoostNet is significantly better than all other baseline models. Its CI (94.43–99.65) shows a statistically significant improvement ($p < 0.001$) compared to other models, with the lower boundary of this CI (94.43) being higher than the upper boundaries of all the competing models. Classical models have large CIs due to reduced accuracy and increased variance, while DL models have smaller intervals, but none of them overlap with TempoBoostNet's.

The performance metrics for the TempoBoostNet proposed model on classifying HC and AD cases are shown in the comparison of the class-wise results in Table 6. The model has been able to achieve very high precision, recall and F1-scores in both classes, with a slightly better score for AD (F1-score: 97.2%) as compared to classifying HC (F1-score: 97%). These results demonstrate that TempoBoostNet can separate these two classes well and that they have robustness up to very good generalization and balance in accuracy across categories.

Fig. 5 depicts the Receiver Operating Characteristic (ROC) curve for the TempoBoostNet, demonstrating its capacity to differentiate between AD cases and HC in the test set. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds, resulting in an AUC of 0.975. An AUC value of 0.975 indicates excellent discriminative power well above chance (AUC = 0.5), suggesting the model's potential to identify AD cases with high sensitivity and a low false positive rate. The balanced distribution of the test set (37 HC, 42 AD) further supports the model's credibility as a reliable candidate for AD detection.

To assess the robustness and generalization capability of the TempoBoostNet model, an additional experiment was conducted using 5-fold cross-validation on the dataset. The dataset was randomly divided into five equal subsets. During each iteration, four subsets were used for training, and the fifth subset was used for testing. This process was repeated five times, and the average performance metrics were calculated. The cross-validation results in Table 7 demonstrate that the proposed TempoBoostNet model consistently maintains high performance across different folds, indicating strong generalization capability and robustness despite the moderate dataset size.

One ablation study was conducted to assess the individual contribution of various components of the proposed TempoBoostNet architecture by comparing simplified versions of the model. Specifically, the performance of models was compared with handcrafted features only, deep contextual embeddings only, and models with either the BiLSTM or XGBoost component removed. This analysis helps in understanding the role played by temporal modeling, feature combination, and the hybrid classification approach.

The findings in Table 8 demonstrate that each aspect contributes to improving the overall performance. The BiLSTM models temporal features within speech signals, while XGBoost enhances classification with strong nonlinear decision boundaries. Additionally, the use of Wav2Vec2 contextual embeddings significantly enhances the representation capacity of the feature set. The overall TempoBoostNet design is the most accurate, achieving a peak accuracy of 97.4%, which validates the effectiveness of combining temporal modeling, feature fusion, and gradient boosting.

A detailed assessment of the TempoBoostNet architecture with current speech-based AD detection strategies is presented in Table 9, based on existing literature. While existing models show promising results, they have limitations such as relying solely on handcrafted acoustic features and classical ML classifiers, which may not adequately capture long-range temporal dynamics and could overfit on small datasets. In contrast, TempoBoostNet addresses these limitations by incorporating deep contextual embeddings from Wav2Vec2.0, utilizing a bidirectional LSTM module to model long-range temporal dependencies and subtle

Table 5
Statistical Significance Test Results.

Models	Accuracy (%)	95% CI (Lower-Upper)
SVM	72.3	63.91 – 80.73
RF	75.1	66.98 – 83.24
LR	70.5	61.95 – 79.07
DT	68.9	60.21 – 77.61
XGBoost	86.5	82.76 – 94.64
DNN	89.2	83.38 – 95.04
CNN	91.5	86.26 – 96.75
LSTM	72.4	63.36 – 80.26
BiLSTM	80.0	72.51 – 87.52
Proposed TempoBoostNet	97.4	94.43 – 99.65

Table 6
Evaluation of TempoBoostNet Model for HC and AD Detection.

Class	Precision (%)	Recall (%)	F1-Score (%)
HC	97.1	96.9	97.0
AD	97.3	97.1	97.2

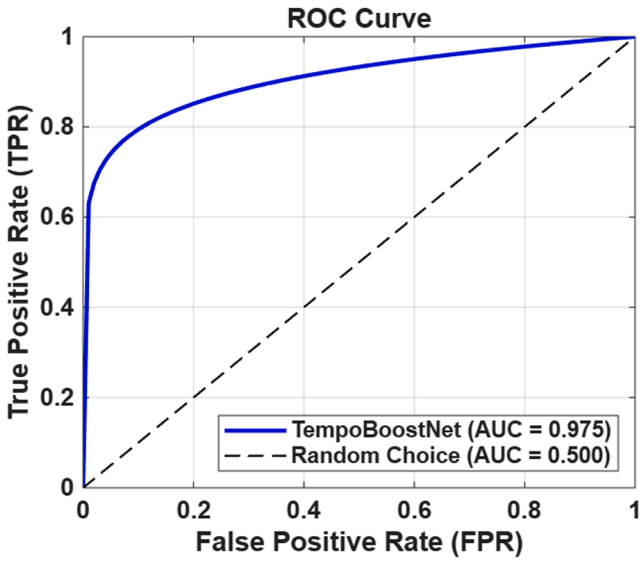


Fig. 5. ROC Curve for TempoBoostNet Model.

Table 7
5-fold Cross-Validation Performance of TempoBoostNet.

Fold	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
Fold 1	96.8	96.5	96.6	96.9
Fold 2	97.2	97.0	97.1	97.3
Fold 3	97.4	97.2	97.3	97.5
Fold 4	96.9	96.7	96.8	97.0
Fold 5	97.3	96.9	97.2	97.4

Table 8
Ablation Study of TempoBoostNet Components.

Model configurations	Accuracy (%)
XGBoost with handcrafted features	86.5
BiLSTM with handcrafted features	80.0
BiLSTM + XGBoost without Wav2Vec2 embedding	93.8
TempoBoostNet (BiLSTM + XGBoost + Wav2Vec2 embedding)	97.4

Table 9
Comparison of Existing and Proposed TempoBoostNet Model for AD Detection.

Ref. No.	Models	Dataset Used	Accuracy (%)
[13]	XGBoost	DementiaBank's Pitt Corpus	75.59
[14]	Stacking classifier	Dementia Pitt corpus repository	97.00
[16]	CNN-LSTM	DementiaBank	72.92
[17]	LR + PCA	DementiaBank dataset	85.00
[18]	SVM	ADReSS-M challenge dataset	69.57
[21]	BERT + ViT + Crossmodal attention	ADReSS-M challenge dataset	91.25
[22]	DenseNet121	MFCC-based speech data	90.00
[23]	DNN	DementiaBank's Pitt Corpus	82.00
[24]	PEADD	Audio-based AD dataset	92.44
Proposed	**TempoBoostNet**	Federated Alzheimer Speech Dataset + Alzheimer Speech Corpus	**97.40**

prosodic variations specific to AD, and leveraging XGBoost for effective regularization and nonlinear discriminative ability on learned sequence representations. This approach results in a significant performance improvement, with a 97.4% accuracy. These findings demonstrate enhanced robustness, generalization, and clinical reliability in the early, non-invasive screening of AD using speech.

5. Conclusion and future work

This paper describes TempoBoostNet, a BiLSTM-XGBoost architecture that recognizes early noninvasive AD in voice recordings. The architecture was designed with noise reduction, silence removal, and an extensive range of handcrafted acoustic-prosodic features, as well as deep contextual embeddings using Wav2Vec2.0; therefore, it acquires nonlinear and temporal features of speech in terms of cognitive decline. The experimental results revealed that TempoBoostNet exhibits significant performance over conventional ML and stand-alone DL baselines, with an accuracy of 97.4%, which is better than the others, proving its strength and stability. The findings highlight the high potential of speech-based AI systems as cost-effective tools that can be easily scaled up to early detect AD. However, a few key steps are required before the translation of these findings to routine clinical care can be conducted. To begin with, the corpora should be expanded to include more age groups, languages, and contexts of speech, which will increase the generalizability and decrease the bias posed by demographic factors. Multimodal extensions, which combine speech and textual information, facial expression analysis, or physiological measurements, can be explored in prospective studies to further improve the accuracy of the diagnosis.

Ethical approval

This material is the authors' own original work, which has not been previously published elsewhere. The paper is not currently being considered for publication elsewhere. The paper reflects the authors' own research and analysis in a truthful and complete manner.

CRediT authorship contribution statement

K. Karthika: . **Anjana Rajesh:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Funding

The authors did not receive support from any organization for the submitted work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Dubois B, Hampel H, Feldman HH, Scheltens P, Aisen P, Andrieu S, et al. Preclinical Alzheimer's disease: definition, natural history, and diagnostic criteria. Alzheimers Dement 2016;12(3):292–323.
[2] Jack Jr CR, Bennett DA, Blennow K, Carrillo MC, Dunn B, Haeberlein SB, et al. NIA-AA research framework: toward a biological definition of Alzheimer's disease. Alzheimers Dement 2018;14(4):535–62.
[3] Fraser KC, Meltzer JA, Rudzicz F. Linguistic features identify Alzheimer's disease in narrative speech. J Alzheimers Dis 2015;49(2):407–22.
[4] Martínez-Nicolás I, Llorente TE, Martínez-Sánchez F, Meilán JJG. Ten years of research on automatic voice and speech analysis of people with Alzheimer's disease and mild cognitive impairment: a systematic review article. Front Psychol 2021;12:620251.
[5] Karthika K. Hierarchical Age and Gender Classification from Speech using Deep Feature Fusion and Enhanced Dimensionality Projection. International Journal of Intelligent Engineering & Systems 2025;18(10):949–63.
[6] De la Fuente Garcia S, Ritchie CW, Luz S. Artificial intelligence, speech, and language processing approaches to monitoring Alzheimer's disease: a systematic review. J Alzheimers Dis 2020;78(4):1547–74.
[7] Ravi, V., EA, G., & KP, S. (2024). Deep learning-based approach for multi-stage diagnosis of Alzheimer's disease. *Multimedia Tools and Applications*, *83*(6), 16799-16822.
[8] Reeha, S., Basavadeepthi, H. M., & Thakur, A. (2023). Alzheimers disease detection using MIC and MLP. In *IEEE 14th International Conference on Computing Communication and Networking Technologies*, pp. 1-6.
[9] Ding K, Chetty M, Noori Hoshyar A, Bhattacharya T, Klein B. Speech based detection of Alzheimer's disease: a survey of AI techniques, datasets and challenges. Artif Intell Rev 2024;57(12):325.
[10] Dinesan, K., Sreesankar, S., VV, D. K., Abhishek, S., & Anjali, T. (2023). Navigating the brain: unveiling alzheimer's disease using convolutional neural network. In *IEEE 7th International Conference on Electronics, Communication and Aerospace Technology*, pp. 763-770.
[11] Kumar MR, Vekkot S, Lalitha S, Gupta D, Govindraj VJ, Shaukat K, et al. Dementia detection from speech using machine learning and deep learning architectures. Sensors 2022;22(23):9311.
[12] Nambiar, A. S., Likhita, K., Pujya, K. S., Gupta, D., Vekkot, S., & Lalitha, S. (2022). Comparative study of deep classifiers for early dementia detection using speech transcripts. In *IEEE 19th India Council International Conference*, pp. 1-6.
[13] Tripathi T, Kumar R. Speech-based detection of multi-class Alzheimer's disease classification using machine learning. International Journal of Data Science and Analytics 2024;18(1):83–96.
[14] Begam RB, Palanivelan M. A speech based diagnostic method for Alzheimer disease using machine learning. Int J Speech Technol 2023;26(4):859–67.
[15] Ravi, V., EA, G., & KP, S. (2024). Deep learning-based approach for multi-stage diagnosis of Alzheimer's disease. *Multimedia Tools and Applications*, *83*(6), 16799-16822.
[16] Mahajan P, Baths V. Acoustic and language based deep learning approaches for Alzheimer's dementia detection from spontaneous speech. Front Aging Neurosci 2021;13:623607.
[17] Shah Z, Sawalha J, Tasnim M, Qi SA, Stroulia E, Greiner R. Learning language and acoustic models for identifying Alzheimer's dementia from speech. Frontiers in Computer Science 2021;3:624659.
[18] Chen, X., Pu, Y., Li, J., & Zhang, W. Q. (2023). Cross-lingual Alzheimer's disease detection based on paralinguistic and pre-trained features. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1-2.
[19] Shah, Z., Qi, S. A., Wang, F., Farrokh, M., Tasnim, M., Stroulia, E., ... & Katsamanis, A. (2023). Exploring language-agnostic speech representations using domain knowledge for detecting Alzheimer's dementia. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1-2.
[20] Burke E, Gunstad J, Pavlenko O, Hamrick P. Distinguishable features of spontaneous speech in Alzheimer's clinical syndrome and healthy controls. Aging Neuropsychol Cogn 2024;31(3):575–86.
[21] Ilias L, Askounis D, Psarras J. Detecting dementia from speech and transcripts using transformers. Comput Speech Lang 2023;79:101485.
[22] Ahn K, Cho M, Kim SW, Lee KE, Song Y, Yoo S, et al. Deep learning of speech data for early detection of alzheimer's disease in the elderly. Bioengineering 2023;10(9):1093.
[23] Park CY, Kim M, Shim Y, Ryoo N, Choi H, Jeong HT, et al. Harnessing the power of voice: a deep neural network model for Alzheimer's disease detection. Dementia and Neurocognitive Disorders 2024;23(1):1.
[24] Xu F, Zheng Q, Shi J, Yan K, Wang M. Pre-training and ensembling based Alzheimer's disease detection. Technol Health Care 2024;32(1):379–95.
[25] Priyanka G, Amshakala K. Predicting dementia through audio: ensemble and deep learning approaches using acoustic features. Comput Biol Med 2025;197:111078.
[26] Kadhim MN, Al-Shammary D, Sufi F. A novel voice classification based on Gower distance for Parkinson disease detection. Int J Med Inf 2024;191:105583.
[27] Yuan L, Liu Y, Feng HM. Parkinson disease prediction using machine learning-based features from speech signal. SOCA 2024;18(1):101–7.
[28] Alghamdi NS, Zakariah M, Karamti H. A deep CNN-based acoustic model for the identification of lung diseases utilizing extracted MFCC features from respiratory sounds. Multimed Tools Appl 2024;83(35):82871–903.

[29] Roy A, Satija U. A novel multi-head self-organized operational neural network architecture for chronic obstructive pulmonary disease detection using lung sounds. IEEE/ACM Trans Audio Speech Lang Process 2024;32:2566–75.
[30] Pradhan JD, Prasad LN, Dash TK, Guduri M, Panda G. Cascaded PFLANN model for intelligent health informatics in detection of respiratory diseases from speech using bio-inspired computation. Journal of Artificial Intelligence and Technology 2024;4 (2):124–31.
[31] La Quatra, M., Orozco-Arroyave, J. R., & Siniscalchi, M. S. (2025). Bilingual dual-head deep model for parkinson's disease detection from speech. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1-5.
[32] https://www.kaggle.com/datasets/programmer3/federated-alzheimers-speech-dataset.
[33] https://github.com/42bismuth/Alzheimer-Detection.